

Exercise 2: UART between two MCUs

Set up UART communication between Arduino UNO / NodeMCU and communicate from one to another. Send for example "Hello" from NodeMCU to Arduino Uno. You control one MCU, your group partner the other.

- Use Serial.begin to start serial communication. Choose same baud rate.
- Make one unit send, and another receive.
How many wires do you need?
Where are the hardware serial pins on NodeMCU/Arduino UNO?
- How do you connect TX/RX? TX -> RX or TX -> TX. Why?
- Will you use *hardware serial* on NodeMCU/Arduino UNO?
- Do you need to use *software serial*?
- Can you also do *full duplex*, or only *half duplex*?
- Can we do **USART** also? What would be needed?

Suggested functions

```
Serial.begin  
Serial.println()  
Serial.print()  
Serial.write()  
Serial.read()  
SoftwareSerial SWserial(RX, TX);  
Serial.readStringUntil()  
while (!Serial) { }
```

